

Sentry 2 Vision Sensor

Manual Sheets

August 08, 2025

The **Sentry 2 Vision Sensor** is programmed to recognize visual features or objects. It is trained to differentiate between a “**Good**” product and a “**Reject**” based on image data. The sensor communicates with the Arduino Nano using either UART (serial) or I2C protocol. The Arduino Nano receives the object classification result (e.g., ID 1 = Good, ID 2 = Reject). *(Ang Sentry 2 Vision Sensor ay naka-program upang makilala ang mga visual na katangian o bagay. Ito ay sinanay upang maiba ang pagitan ng “Good” na produkto at “Reject” na produkto batay sa image data. Nakikipag-ugnayan ang sensor sa Arduino Nano gamit ang UART (serial) o I2C protocol. Tinatanggap ng Arduino Nano ang resulta ng classification ng object (halimbawa: ID 1 = Good, ID 2 = Reject).)*

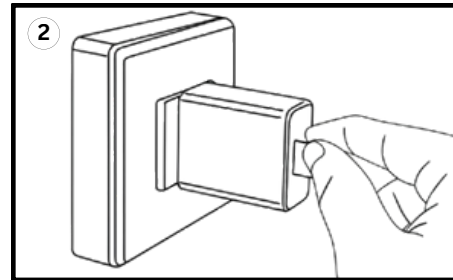
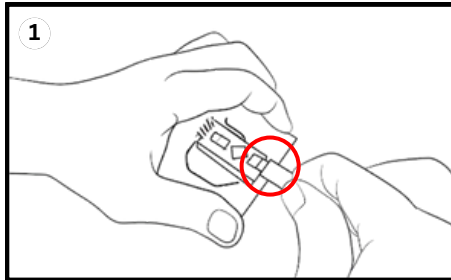
- **How does it work? (Paano ito gumagana?)**

1. Training the Sentry 2 Vision Sensor: *(Pagsasanay sa Sentry 2 Vision Sensor:)*
 - a. Before using, the Sentry 2 Vision Sensor is trained to recognize products. *(Bago gamitin, sinasanay muna ang Sentry 2 Vision Sensor upang makilala ang mga produkto.)*
 - b. For example: *(Halimbawa:)*
 - Object 1 (ID: 1) = Good product
 - Object 2 (ID: 2) = Reject product
 - c. This training is done using a button on the sensor or using software on a PC. *(Ang pagsasanay na ito ay ginagawa gamit ang button sa sensor o gamit ang software sa PC.)*
2. Detecting the product: *(Pag-detect ng produkto:)*
 - a. When a product passes in front of the sensor, the Sentry 2 takes a picture and compares it with the trained objects. *(Kapag may produktong dumaan sa harap ng sensor, kinukuhanan ito ng larawan ng Sentry 2 at inihahambing sa mga naunang sinanay na object.)*
 - b. It identifies whether it matches the Good or Reject product. *(Tinutukoy nito kung tumutugma ito sa Good o Reject na produkto.)*

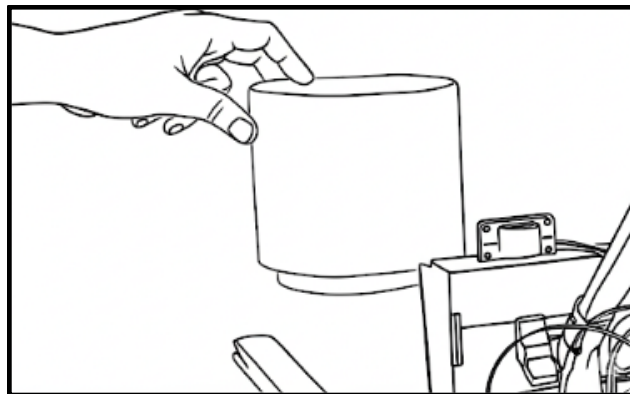
3. Sending Data to Arduino Nano: *(Pagpapadala ng Data sa Arduino Nano:)*
 - a. After recognizing the object, the sensor sends the result (Object ID) to the Arduino Nano using UART or I2C communication. *(Matapos makilala ang object, ipinapadala ng sensor ang resulta (Object ID) sa Arduino Nano gamit ang UART o I2C na komunikasyon.)*
4. Arduino Decides: *(Pagdedesisyon ng Arduino:)*
 - a. The Arduino Nano reads the object ID. *(Binabasa ng Arduino Nano ang Object ID.)*
 - b. It checks: *(Tinitingnan nito kung:)*
 - If ID == 1 → Product is Good
 - If ID == 2 → Product is Reject
5. Displaying the Result on I2C LCD *(Pagpapakita ng Resulta sa I2C LCD)*

How to use the device

STEP 1: Plug the DEVICE to the power supply. (*Isaksak ang DEVICE sa power supply.*)



STEP 2: Place products one by one in front of the sensor. (*Ilagay ang mga produkto isa-isa sa harap ng sensor.*)

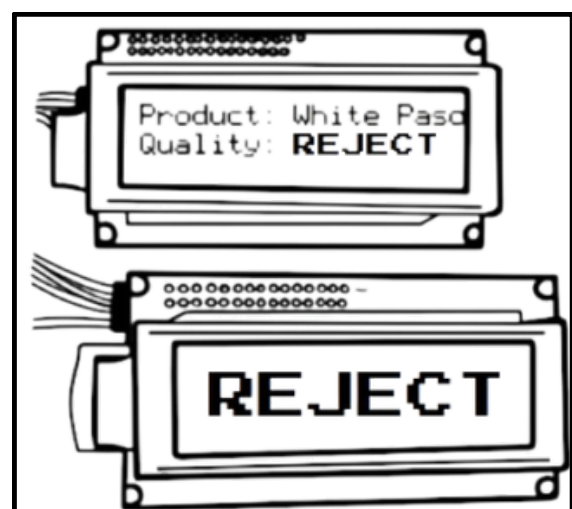


Wait for a few seconds to scan the product and the result will show on the LCD: (*Mag-antay ng ilang segundo sa pag-iscan ng produkto.*)

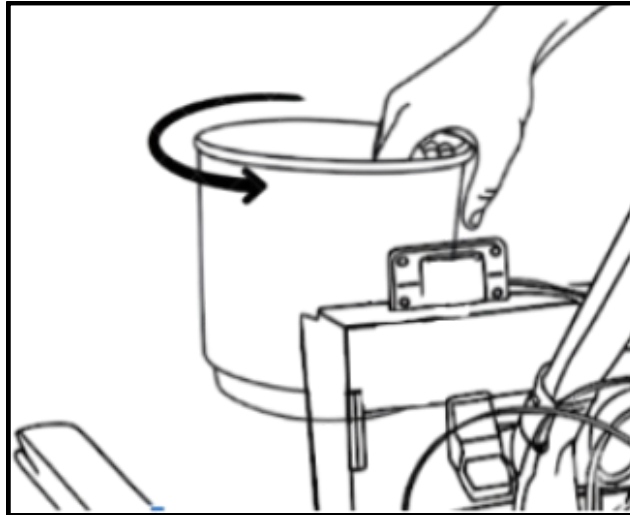
a: GOOD = Pass



b: REJECT = Fail

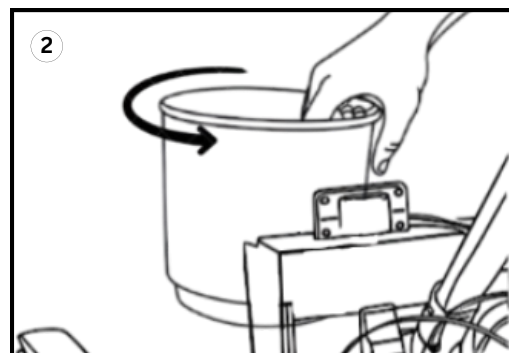


STEP 3: Rotate the product on four different angles to see the defects on the different parts of the product. (*Ikutin ang produkto sa apat na magkakaibang anggulo upang makita ang sira o depektibong parte.*)



After rotating the product, wait a few seconds for the scan to complete. The result will then be shown on the LCD. (*Pagkatapos i-rotate ang produkto, maghintay ng ilang segundo para ma-scan ito. Pagkatapos, lalabas ang resulta sa LCD.*)

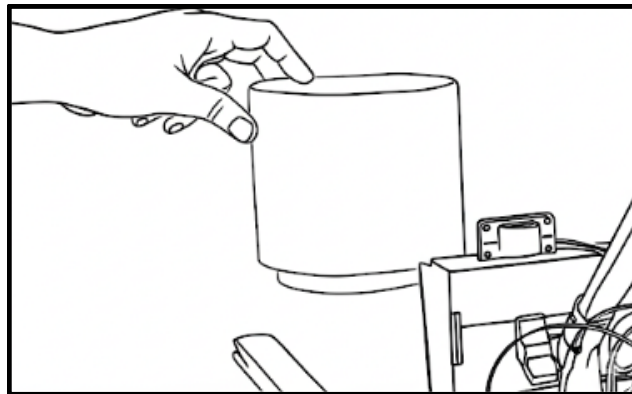
STEP 4: Repeat as needed, the device works continuously as long as it is powered. (*Ulitin lang ang mga proseso sa ibang produkto, ang device ay gumagana nang tuloy-tuloy hangga't mayroong power supply.*)



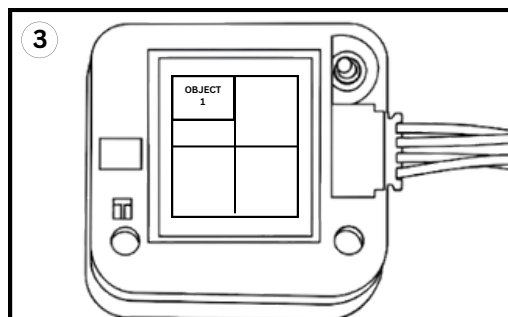
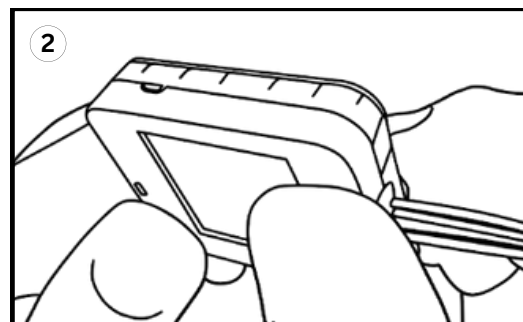
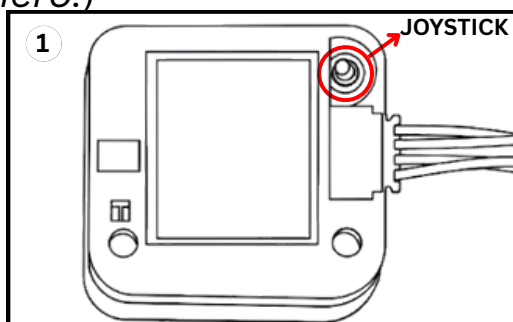
Learning Process

- Teaching GOOD Product -

STEP 1: Put a GOOD PRODUCT in front of the camera. (*Ilagay ang GOOD PRODUCT sa harap ng kamera.*)



STEP 2: Press the JOYSTICK and wait for the Sentry to recognize the product. the sentry will display a box surrounding the object and has a name of object with a corresponding number. (*Pindutin ang JOYSTICK at hintayin na matandaan ng Sentry ang PRODUCT. Magpapakita ang Sentry ng BOX na nakapalibot sa PRODUCT at may pangalan na "OBJECT" na may kaukulang numero.*)

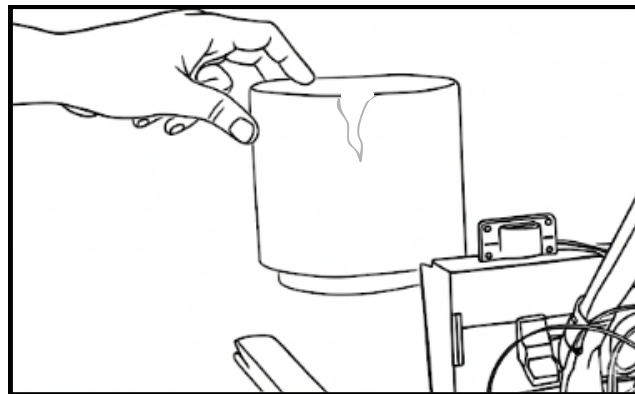


Repeat steps 1 and 2 for six to seven different products

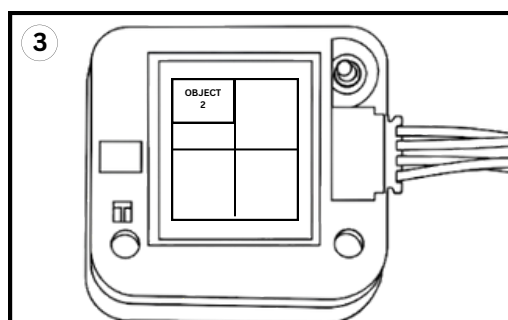
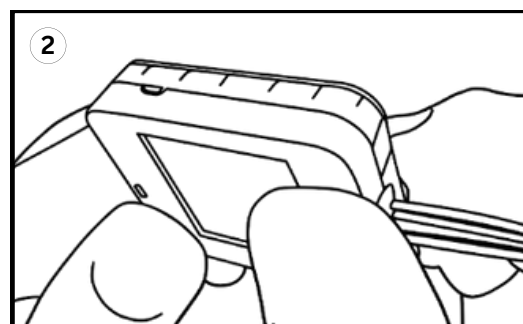
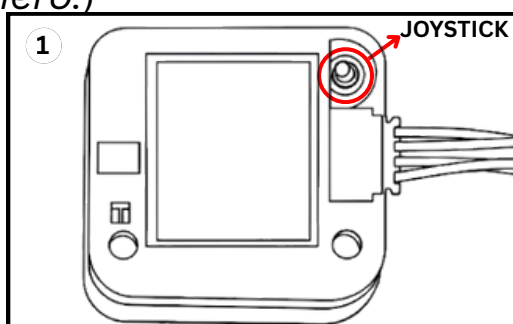
Learning Process

- Teaching REJECT Product -

STEP 1: Put a REJECT PRODUCT in front of the camera. (*Ilagay ang GOOD PRODUCT sa harap ng kamera.*)

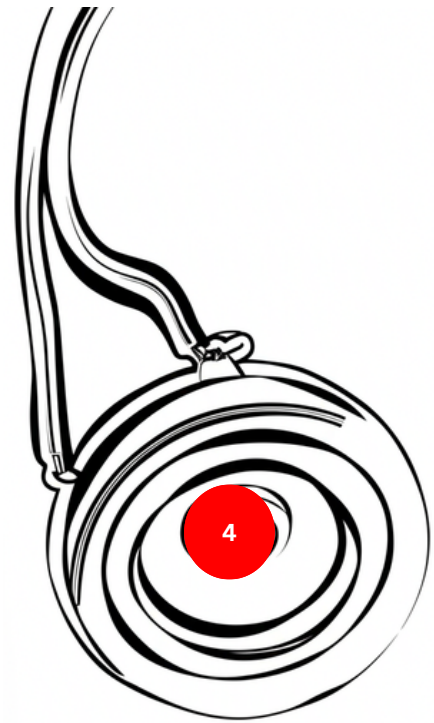
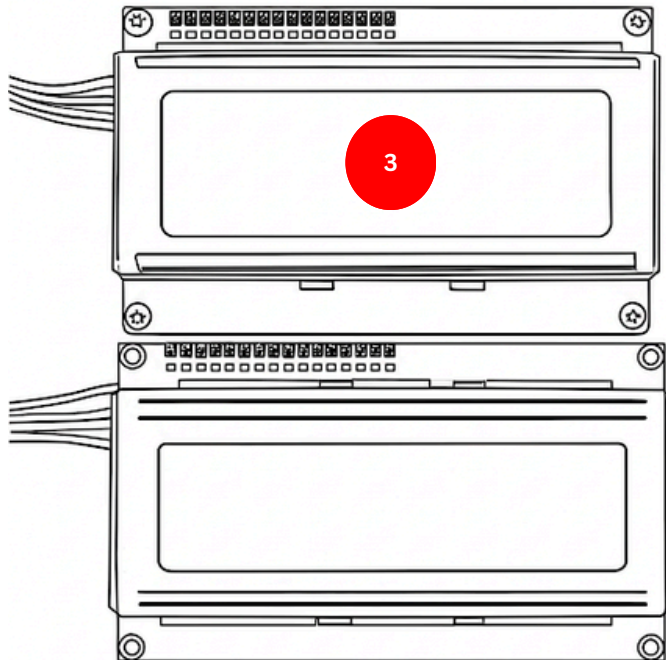
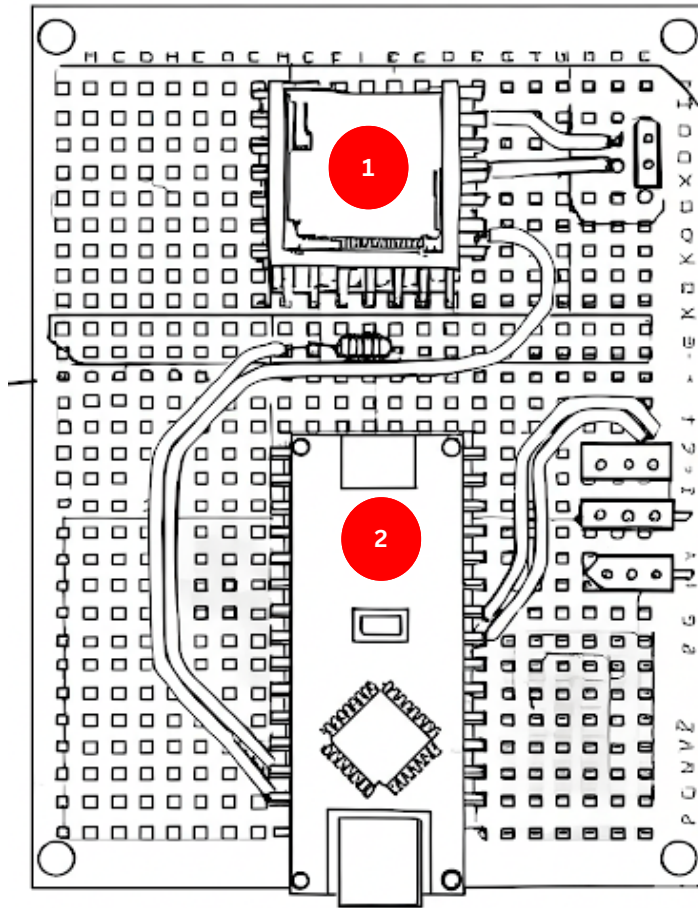


STEP 2: Press the JOYSTICK and wait for the Sentry to recognize the product. the sentry will display a box surrounding the object and has a name of object with a corresponding number. (*Pindutin ang JOYSTICK at hintayin na matandaan ng Sentry ang PRODUCT. Magpapakita ang Sentry ng BOX na nakapalibot sa PRODUCT at may pangalan na "OBJECT" na may kaukulang numero.*)



Repeat steps 1 and 2 for six to seven different products

Hardware Components



Hardware Description

1	DF PLAYER MINI	Used to play audio files stored on a memory card. Provides audio alerts/notifications in the system. Ensure the memory card is formatted to FAT32.
2	ARDUINO NANO	Main microcontroller that processes data from the Sentry2 Vision Sensor and controls the displays, DFPlayer Mini, and speaker.
3	20x4 LCD DISPLAY	Displays product details such as ID, name, and inspection status during quality control. Shows whether the inspected product is Good or Rejected based on the Sentry2 Vision Sensor analysis.
4	4 Ohms SPEAKER	Provides audio output from the DFPlayer Mini for alarms, status updates, or guidance messages.

Sentry 2 Vision Camera Specification

- **Processor:** Built around the K210, a 64-bit RISC-V neural network processor capable of running AI algorithms directly on the device.
- **Camera Resolution:** Captures images at 240×240 pixels, sufficient for basic object detection and recognition tasks.
- **Wireless Connectivity:** Equipped with an ESP8285 Wi-Fi chip, allowing for wireless communication and control.
- **Programming Support:** Compatible with the Arduino IDE, enabling users to write and upload custom code easily.
- **Interfaces:** Supports multiple communication interfaces, including USB, UART, and I2C, facilitating integration with various microcontrollers and development boards.
- **Dimensions:** Compact size of 43mm x 36mm x 12mm, making it easy to incorporate into different projects.
- **User Interaction:** Features a joystick for easy navigation between different vision modes and settings.

Do's and Don'ts in using Sentry 2 Vision Sensor

Do's (Things You Should Do)

- **Connect with USB-C first**

Plug the Sentry 2 into your computer via a USB-C cable to power it and program it.

- **Use the joystick to pick a vision mode**

Move the little joystick up or down to choose the vision type you want (like object tracking, color detection, etc.)

- **Point the camera directly to what you want it to see**

Make sure the sensor is facing the object you want it to recognize—it won't work otherwise.

- **Adjust camera settings for better image quality**

You can tweak settings such as zoom, brightness, white balance, and saturation to get clearer images.

- **Use a supported micro-SD (TF) card for storage**

You can store images on a micro-SD card—just make sure it's one that works with SPI mode.

- **Be careful when using power via USB and the 4-pin port simultaneously**

If you power the Sentry 2 through USB, do not also power it via the 4-pin port—it may cause damage.

Don'ts (Things to Avoid)

- **Don't leave the camera facing nothing**

If there's nothing in view, it can't recognize anything. Always point the camera at something you want to track.

- **Don't use an unsupported SD card**

Some SD cards don't work well with SPI mode. Using one that's not supported may cause errors or it simply won't store images.

- **Don't power it two ways at once**

Avoid using USB power and external power through the 4-pin connector at the same time because it could damage the sensor.

Arduino Code Manual

Manual Sheets

Code Overview

- Sentry2 for object detection/classification
- Two I2C LCD's (20x4) for visual feedback
- DFPlayer Mini for audio alerts
- State Machine logic to handle product inspection workflow

Set-ups

Download this link for the code of the Arduino:

https://drive.google.com/drive/folders/1fNTrzEJZCZKdlqPH7WJjseKyj74osB0F?usp=drive_link

QR CODE:



Key Constants and Variables

- Global Settings:

```
#define VISION_TYPE Sentry::kVisionLearning // Sentry2 mode
#define MIN_DETECTION_CONFIDENCE 3 // Minimum detection count for confirmation
#define NO_DETECTION_THRESHOLD 5 // Frames to return to scanning
// Audio file numbers (upload these to your DFPlayer Mini SD card)
#define REJECT_SOUND 1 // SD: 001.mp3
#define GOOD_SOUND 2 // SD: 002.mp3
```

- Product Database:

```
// Product definitions with fixed reject types
```

```
struct Product {
    int id;
    const char* name;
    const char* quality; // "GOOD" or "REJECT"
    const char* rejectReason; // Fixed reject reason
};
```

```
Product products[] = {
    {0, "Unknown", "UNKNOWN", "Unknown"},
    {1, "White Paso", "GOOD", "N/A"},
    {2, "White Paso", "REJECT", "Cracking"},
    {3, "White Paso", "REJECT", "Cracking"},
    {4, "Black Paso", "GOOD", "N/A"},
    {5, "Black Paso", "REJECT", "Cracking"},
    {6, "Footing", "REJECT", "Moisture"},
    {7, "Footing", "GOOD", "N/A"},
    {8, "Core Clear", "REJECT", "Black Spot"},
    {9, "Core Clear", "GOOD", "N/A"},
};
```

- System States:

```
// Display states
enum DisplayState {
    STATE_INIT,
    STATE_DETECTING,
    STATE_SHOW_GOOD,
    STATE_SHOW_REJECT
};
```

Critical Functions

1. setup()

- Initializes hardware and states:
- Starts serial communication
- Initializes DFPlayer (halts on failure)
- Configures LCD displays
- Initializes Sentry2 vision sensor
- Sets initial state to STATE_DETECTING

2. updateProductDetails(int label)

- Handles product results:
- Plays GOOD/REJECT sound via DFPlayer
- Updates details LCD with:
 - Product name
 - Quality status
 - Reject reason (if applicable)

3. displayGood() / displayReject()

- Renders large-font "GOOD" or "REJECT" on main LCD using big font library.

4. loop()

Core state machine logic:

- Checks Sentry2 for detected objects
- Requires MIN_DETECTION_CONFIDENCE consistent readings
- Updates displays and plays sound on confirmation
- Resets to scanning after NO_DETECTION_THRESHOLD empty cycles

-

Customization Guide

- Modify the products:
 - Edit the products[] array to match your trained Sentry2 labels:

```
{ID, "Product Name", "QUALITY", "Reject Reason"}
```

- Adjust Sensitivity:

```
// Increase for stricter detection  
#define MIN_DETECTION_CONFIDENCE 5
```

```
// Decrease for faster reset  
#define NO_DETECTION_THRESHOLD 3
```

- Changed Audio Files:
 - Update SD card contents and redefine:

```
#define REJECT_SOUND 5 // Now plays 005.mp3  
#define GOOD_SOUND 6 // Now plays 006.mp3
```

Adding new products on Sentry 2 Vision Sensor

- **Update Product Database in Code**

1. Locate this product array: *(Hanapin ang product array na ito:)*

- Product products[] = {
- {0, "Unknown", "UNKNOWN", "Unknown"},
- {1, "White Paso", "REJECT", "Cracking"},
- {2, "White Paso", "GOOD", "N/A"},
- // ... existing products
- };

2. Add a new line with your product information: *(Magdagdag ng bagong line para sa impormasyon ng iyong produkto:)*

{ID, "Product Name", "GOOD or REJECT", "Reject Reason or N/A"},

Example:

{11, "Red Cap", "GOOD", "N/A"},

{12, "Red Cap", "REJECT", "Scratches"},

3. Compile and upload the modified code to your Arduino. *(I-compile at i-upload ang binagong code sa iyong Arduino.)*

Physical Setup for Learning

Prepare Learning Environment such as: *(Ihandang mga sumusunod:)*

Lighting: Ensure consistent, bright lighting *(Tiyaking maliwanag ang ilaw.)*

Background: Use a plain, contrasting background *(Gumamit ng simpleng background.)*

Camera Position: Keep Sentry 2 at fixed distance and angle. *(Panatilihin ang Sentry 2 sa nakatakdang distansya at anggulo.)*

Sample Products: Have both GOOD and REJECT samples ready. *(Ihandang REJECT at GOOD products.)*

Product Positioning: *(Posisyon ng produkto:)*

- Place product in center of Sentry 2's field of view *(Ilagay ang produkto sa tapat ng Sentry 2.)*
- Ensure product is clearly visible and in focus *(Siguraduhing nakatapat ang sentry 2 at nakafocus sa produkto)*
- Avoid shadows or reflections on the product *(Iwasan ang mga anino o kaya naman repleksyon na tumama sa produkto)*

Troubleshooting

Common Issues:

1. No LCD Output:

- Verify I2C addresses with I2C scanner sketch
- Check begin() calls in setup

2. Incorrect Detections:

- Ensure Sentry2 labels match products[] IDs
- Increase MIN_DETECTION_CONFIDENCE

3. No Audio:

- Confirm files are named 001.mp3, 002.mp3, etc.
- Check DFPlayer RX/TX connections (Pins 10/11)

4. Sentry2 Failures:

- Check I2C wiring (SDA/SCL to A4/A5)
- Verify sufficient lighting

Key Dependencies

Required Libraries:

1. Wire.h (Built-in)
2. LiquidCrystal_I2C.h
3. BigFont01_I2C.h
4. Sentry.h
5. DFRobotDFPlayerMini.h

Install via Arduino IDE:

Sketch > Include Library > Manage Libraries > Search names

Download Sentry2 library:

<https://github.com/AlTosee/Sentry-Arduino/releases>

For more information and extended functionality, see Sentry documentation at:

<https://tosee.readthedocs.io/en/latest/Sentry2/introduction.html>